

3.5 Integration

Pure Mathematics 3 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Advanced methods include partial fractions, substitution and parts, and trig/exponential integrals. For integrals with a linear inner function multiply by reciprocal of its derivative. Improper integrals need limiting processes when bounds or integrand are unbounded.

Worked example: follow the reasoning

$$\int (2x+1)^{-1} dx = (1/2) \ln |2x+1| + C.$$

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Evaluate $\int$ from 0 to 1 of $1/(x+1) dx$.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Antiderivative $\ln(x+1)$ [2]; $\ln 2 - \ln 1 = \ln 2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Mathematics 9709, 2026–2027. <https://www.cambridgeinternational.org/Images/697427-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.